How Economists Discuss Central Bank Digital Currencies

Themes, tone and stance, 2014–2026: a bibliometric and text-as-data study

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October 2026

M2 – Private crypto

Slide 1 **Time: 0:15** clock at the end of the slide: **0:15** / 25:55

Good morning. We are Louis Drogo, Kenza Redwane and Adriel Mazere. Our paper asks how economists discuss central bank digital currencies, and how that discussion moved between 2014 and 2026. **Total budget: about 25 minutes, then questions.**



The paper in one slide

Research question. How has economists and institutional discourse on CBDCs evolved between 2014 and 2026, and to what extent does this evolution align with changes in the policy environment?

Who writes?	research economists, central banks and institutions
What do they write about?	themes and topics, and how they shift
How do they discuss CBDCs?	**tone and stance: the core of the paper**
Why, in their own words?	stated motivations
When, and with which environment?	bursts, policy events, sanctions

Scope. Descriptive: we measure changes and show what moves with them; we do not claim causes.

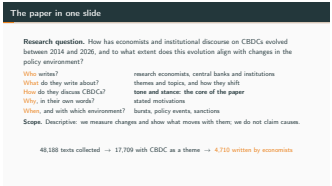
48,188 texts collected $\rightarrow$ 17,709 with CBDC as a theme $\rightarrow$ 4,710 written by economists

Slide 2 **Time: 1:00** clock at the end of the slide: **1:15** / 25:55

Say: Read the research question. Then: we split it into five questions, the five Ws.

- Who writes, what they write about, **how** they discuss CBDCs (tone and stance, the core), why (their stated motivations), and when (the policy environment).
- Scope: descriptive. We measure how the discussion changes and what moves with it; we do not claim causes.
- The funnel: 48,188 texts collected, 17,709 with CBDC as a theme, 4,710 written by economists. Everything that follows is on those 4,710.

If asked "why not causal?": no exogenous variation for most questions; where there is some (events, sanctions) we test it, with placebo dates and a difference-in-differences, at the end.



Outline

Introduction

Data

Who writes

What is written

How: tone and stance

Why: stated motivations

When: the environment

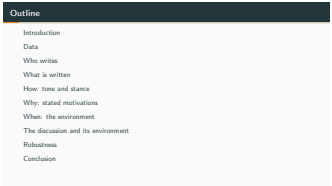
The discussion and its environment

Robustness

Conclusion

Slide 3 **Time: 0:10** clock at the end of the slide: **1:25** / 25:55

One sentence: introduction and data, then one section per W, then how the discussion lines up with its environment, robustness and conclusion.



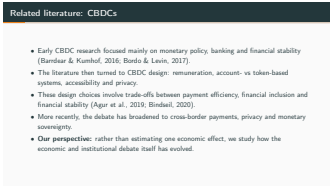
Introduction

Related literature: CBDCs

- Early CBDC research focused mainly on monetary policy, banking and financial stability (Barrdear & Kumhof, 2016; Bordo & Levin, 2017).
- The literature then turned to CBDC design: remuneration, account- vs token-based systems, accessibility and privacy.
- These design choices involve trade-offs between payment efficiency, financial inclusion and financial stability (Agur et al., 2019; Bindseil, 2020).
- More recently, the debate has broadened to cross-border payments, privacy and monetary sovereignty.
- **Our perspective:** rather than estimating one economic effect, we study how the economic and institutional debate itself has evolved.

Slide 4 **Time: 0:45** clock at the end of the slide: **2:10** / 25:55

Say: The economics of CBDCs started with monetary policy and banking: Barrdear and Kumhof (2016), Bordo and Levin (2017). Then design questions: interest-bearing or not, accounts or tokens, privacy; with trade-offs between efficiency, inclusion and stability (Agur, Ari and Dell'Ariccia 2019; Bindseil 2020 on tiered remuneration). More recently: cross-border payments, privacy, monetary sovereignty. **Our angle:** we do not estimate one effect of CBDCs; we measure how the debate itself has evolved.



Related literature: what the field builds on

- **Privacy vs. state-issued digital money:** digital payment systems raise the question of whether privacy can be preserved without relying on traditional intermediaries.
- **Financial efficiency vs. financial stability:** safer and more efficient public money may also draw funds away from commercial banks and affect credit provision.
- **Monetary sovereignty in the digital age:** cryptocurrencies and stablecoins raise questions about the coexistence of public and private digital money.

$\Rightarrow$ These tensions increasingly connect CBDCs to cross-border and geopolitical issues.

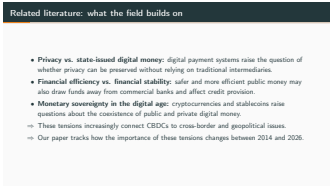
$\Rightarrow$ Our paper tracks how the importance of these tensions changes between 2014 and 2026.

Slide 5 **Time: 0:45** clock at the end of the slide: **2:55** / 25:55

Say: Three tensions predate CBDCs.

- Privacy against state money: Chaum's (1983) blind signatures, Bitcoin (2008).
- Efficiency against stability: banks transform deposits into credit (Diamond and Dybvig 1983); safe public money could drain deposits (Andolfatto 2020; Brunnermeier and Niepelt 2019).
- Sovereignty: stablecoins and global private money.

These tensions now meet cross-border and geopolitical issues. We track how much weight each gets over time; you will see them again as motivations.



What this paper adds to 29 earlier reviews

Five recent bibliometric reviews:

Study	Data	Documents	Tools
Kvedaravičiūtė & Šapkauskienė (2024)	WoS + Scopus	228	VOSviewer
Fahad & Bulut (2024)	Scopus	169 of 323	Biblioshiny
Bapat & Sharma (2025)	Scopus	371	bibliometrics + TCCM
Prodan et al. (2024)	Scopus + WoS	747 of 1,024	PRISMA, VOSviewer
Paulovici (2026)	Scopus	892	VOSviewer
This study	**5 databases + central banks**	**4,710**	**+ text as data**

- Gap: one or two databases, mostly English journal articles; central banks absent; who writes is not checked.
- Maps say *what* is studied, not *how* (tone, stance) or *why*; no measure is validated.
- Here: economists only (checked text by text), all languages, tone and stance over time, every coder validated.

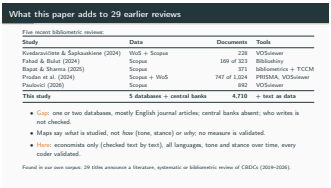
Found in our own corpus: 29 titles announce a literature, systematic or bibliometric review of CBDCs (2019–2026).

Slide 6 **Time: 1:00** clock at the end of the slide: **3:55** / 25:55

Say: Our own corpus contains 29 reviews of the CBDC literature. The five most recent bibliometric ones share three limits:

- one or two databases (Scopus, Web of Science), mostly English journal articles, central banks absent;
- maps of *what* is studied, not *how* (tone, stance) or *why*;
- no measure is checked against a reading of the texts.

We use five databases plus central banks, all languages, check that each text is by an economist, and validate every coder. **If asked about the closest paper**: Hoang, Ngo and Vu (2023, RIBAF) do text mining, but on academic papers only and without stance.



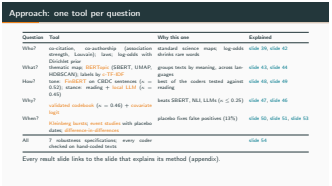
Approach: one tool per question

Question	Tool	Why this one	Explained
Who?	co-citation, co-authorship (association strength, Louvain); laws; log-odds with Dirichlet prior	standard science maps; log-odds shrinks rare words	slide 39, slide 42
What?	thematic map; BERTopic (SBERT, UMAP, HDBSCAN); labels by c-TF-IDF	groups texts by meaning, across languages	slide 43, slide 44
How?	tone: FinBERT on CBDC sentences ($\kappa = 0.52$); stance: reading + local LLM ($\kappa = 0.45$)	best of the coders tested against reading	slide 48, slide 49
Why?	validated codebook ($\kappa = 0.46$) + covariate logit	beats SBERT, NLI, LLMs ($\kappa \leq 0.25$)	slide 47, slide 46
When?	Kleinberg bursts; event studies with placebo dates; difference-in-differences	placebo fixes false positives (13%)	slide 50, slide 51, slide 53
All	7 robustness specifications; every coder checked on hand-coded texts		slide 54

Every result slide links to the slide that explains its method (appendix).

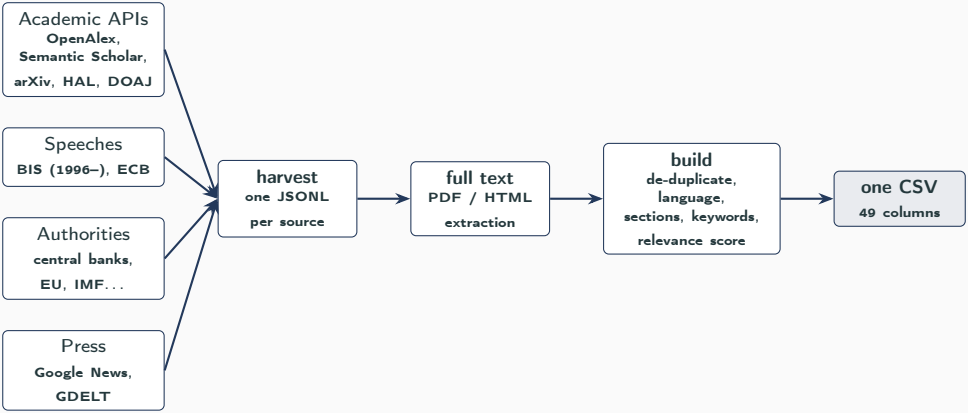
Slide 7 **Time: 0:45** clock at the end of the slide: **4:40** / 25:55

Say: One tool per question. Who: networks and bibliometric laws, plus a word-level comparison of groups. What: thematic map and a topic model that works on meaning, across languages. How: FinBERT for tone, reading plus a local language model for stance. Why: a validated codebook. When: bursts, event studies with placebo dates, a difference-in-differences. Each automatic coder was compared with our reading; the agreement (kappa) is shown. Methods are in the appendix, linked from every slide.



Data

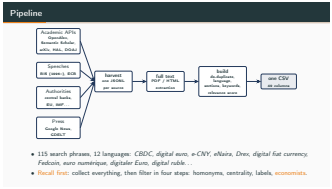
Pipeline



- 115 search phrases, 12 languages: *CBDC, digital euro, e-CNY, eNaira, Drex, digital fiat currency, Fedcoin, euro numérique, digitaler Euro, digital ruble...*
- Recall first: collect everything, then filter in four steps: homonyms, centrality, labels, economists.

Slide 8 **Time: 0:40** clock at the end of the slide: **5:20** / 25:55

Say: Four families of sources: academic databases (OpenAlex, Semantic Scholar, arXiv, HAL, DOAJ), central bank speeches (BIS archive since 1996, ECB), central bank and institution websites, and the press. 115 search phrases in 12 languages: CBDC, digital euro, e-CNY, eNaira, euro numérique, digitaler Euro, digital ruble... Principle: **recall first**. Collect everything, then filter in four steps. The fourth is the one the course asked for: is it written by an economist?



Step 1: is it really *Central Bank Digital Currency*?

```
TERMS = {
  "CBDC": r"\bCBDCs?\b|\bwCBDCs?\b|\brCBDCs?\b",
  "central bank digital currency":
      r"central[\s-]+banks?(?:['']s?)?[\s-]+"
      r"digital[\s-]+currenc(?:y|ies)",
  "digital euro": r"digital(?:e[rn]?)?\s+euro\b"
      r"|euro\s+numérique|euro\s+digitale?\b",
  "DCEP (e-CNY)": r"\bDC/?EP\b",
  ...
}
```

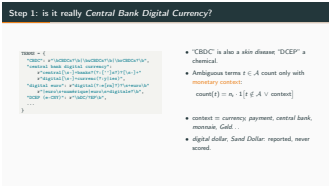
- "CBDC" is also a *skin disease*; "DCEP" a chemical.
- Ambiguous terms $t \in \mathcal{A}$ count only with monetary context:

$$\text{count}(t) = n_t \cdot 1\big[t \notin \mathcal{A} \vee \text{context}\big]$$

- context = *currency, payment, central bank, monnaie, Geld...*
- *digital dollar*, *Sand Dollar*: reported, never scored.

Slide 9 **Time: 0:30** clock at the end of the slide: **5:50** / 25:55

Say: Step 1 removes homonyms. “CBDC” is also a skin disease in medical papers, “DCEP” a chemical. Ambiguous terms only count when the text also talks about money: currency, payment, central bank, in several languages. “Digital dollar” and “Sand Dollar” are recorded but never scored.



Step 2: how central is CBDC to the text?

Each available field f gets a centrality $c_f \in [0, 1]$:

$$c_{\text{title}} = 1[\text{CBDC term in title}] \qquad c_{\text{kw}} = 0.4 \cdot 1[\text{CBDC in keywords}]$$

$$c_{\text{abs}} = \frac{\#\{\text{sentences mentioning CBDC}\}}{\#\{\text{sentences}\}} \qquad c_{\text{body}} = \tfrac{1}{2} \min\left(1, \frac{\rho}{3}\right) + \tfrac{1}{2}\, \sigma$$

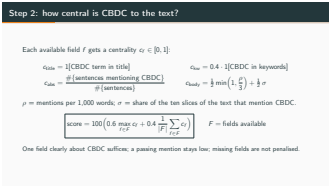
ρ = mentions per 1,000 words; σ = share of the ten slices of the text that mention CBDC.

$$\boxed{\text{score} = 100\left(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\right)} \qquad F = \text{fields available}$$

One field clearly about CBDC suffices; a passing mention stays low; missing fields are not penalised.

Slide 10 **Time: 0:30** clock at the end of the slide: **6:20** / 25:55

Say: Step 2: how central is CBDC to the text, from 0 to 100. Each field gets a centrality: the title (1 if it names a CBDC), the keywords (only 0.4, because databases assign them partly by machine), the abstract (share of its sentences on CBDCs), the body (how dense and how spread out the mentions are). The score is 60% the strongest field and 40% the average: one field clearly about CBDCs is enough, a passing mention stays low, and missing fields do not count against a text. **If asked "why 0.6/0.4?"**: chosen so that a title alone makes a main theme, keywords alone never do, and a passing mention scores about 10; then validated by reading (next slide).



Step 3: labels, and how good is the filter?

Label (all texts kept)	Texts	Theme	95% CI
Main theme (≥ 50)	15,464	29/30	0.83–0.99
Secondary theme (25–49)	2,245	11/29	0.23–0.56
Mentioned only (< 25)	489	1/30	0.01–0.17
Press, text not available	13,446	2/19	0.03–0.31
Ambiguous terms only	1,709	3/10	0.11–0.60
Off-topic (homonyms, no match)	14,803	0/20	0.00–0.16
Dated before 2014 (checked: false)	32	–	–

Theme = share read as having CBDC as a theme, in a stratified sample of 138 texts read blind to the label; Wilson intervals.

Kept: main + secondary theme (17,709 texts).

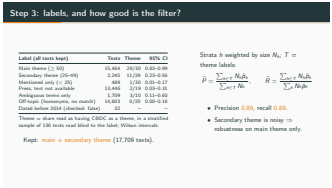
Strata h weighted by size N_h; T = theme labels:

$$\widehat{P} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h \in T} N_h}, \qquad \widehat{R} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h} N_h \hat{p}_h}$$

- Precision 0.89, recall 0.89.
- Secondary theme is noisy $\Rightarrow$ robustness on main theme only.

Slide 11 **Time: 0:40** clock at the end of the slide: **7:00** / 25:55

Say: Step 3: we read 138 texts without seeing their label. 29 of 30 "main theme" texts really are about CBDCs, none of the off-topic ones are. Weighted by the size of each group, the filter has a precision and a recall of 0.89. The "secondary theme" group is noisy (11 of 29), so every result is re-run on main-theme texts only (robustness slide). We keep main and secondary theme: 17,709 texts.



Step 4: is it written by an economist?

Texts with CBDC as a theme	Texts	
Research, classified in economics or finance (OpenAlex)	3,783	kept
Research, other field but an author at a central bank, IMF or BIS	148	kept
Research without OpenAlex record, economics venue	63	kept
Research in other fields	3,469	dropped
Central bank speeches	511	kept
Documents of central banks and international institutions	205	kept
Documents of governments and parliaments	34	dropped
Press articles (no author)	9,496	dropped

Field = OpenAlex primary topic ("Economics, Econometrics and Finance"); institutions = authors' affiliations in OpenAlex. Research dropped: Computer Science 1,769, Business, Management and Accounting 663, no OpenAlex record 473, Social Sciences 281.

- 4,710 texts by economists: research economists, and central banks and international institutions (staff economists, institutional voice).
- Press dropped: no author is known, and journalists are not economists.

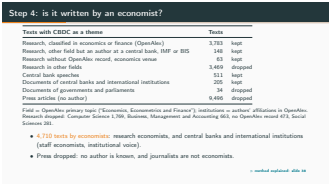
▷ method explained: slide 38

Slide 12 **Time: 0:50** clock at the end of the slide: **7:50** / 25:55

Say: Step 4 answers the course's question: is the text written by an economist?

- Research is kept if OpenAlex classifies it in economics and finance, or if an author works at a central bank, the IMF or the BIS, or, without an OpenAlex record, if it appears in an economics outlet.
- Central bank speeches and documents are kept as institutional voices.
- Dropped: research in other fields (computer science first), governments and parliaments, the press.

Result: **4,710 texts by economists**: about 4,000 research texts, 511 speeches, 205 documents. **If asked "a speech by a lawyer governor?"**: kept, as the institution's voice, written with its economists; the robustness slide also shows the results with every discipline.



Sources and completeness (economists' corpus)

Database	Documents
OpenAlex	4,043
Semantic Scholar	2,164
BIS central bankers' speeches	465
DOAJ	251
Central banks & authorities (crawl)	205
ECB speeches	108
HAL	31
arXiv	24

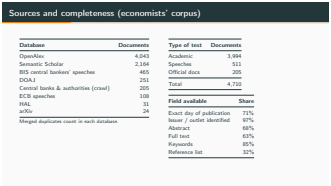
Merged duplicates count in each database.

Type of text	Documents
Academic	3,994
Speeches	511
Official docs	205
Total	4,710

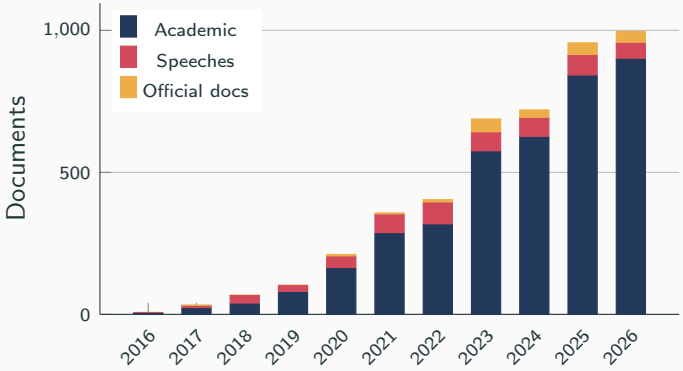
Field available	Share
Exact day of publication	71%
Issuer / outlet identified	97%
Abstract	68%
Full text	63%
Keywords	85%
Reference list	32%

Slide 13 **Time: 0:30** clock at the end of the slide: **8:20** / 25:55

Say: OpenAlex brings most research, the BIS archive most speeches. A text found in two databases is merged and counted in both. Completeness: exact day for 71%, abstract 68%, full text 63%, reference lists only 32%. Each analysis uses the texts that have its input; the number of texts is always on the slide. *If short of time: one sentence.*



A literature that took off after 2020

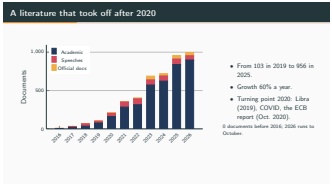


- From 103 in 2019 to 956 in 2025.
- Growth 60% a year.
- Turning point 2020: Libra (2019), COVID, the ECB report (Oct. 2020).

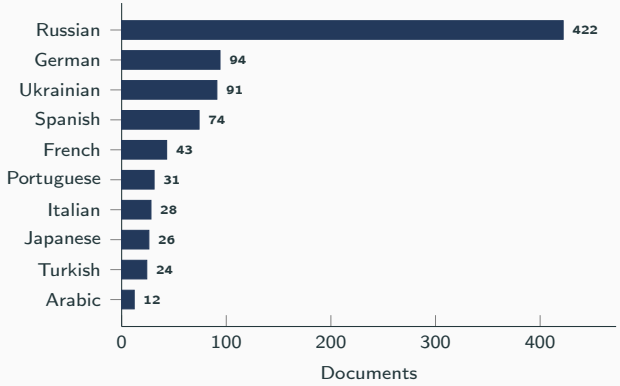
0 documents before 2016; 2026 runs to October.

Slide 14 **Time: 0:30** clock at the end of the slide: **8:50** / 25:55

Say: Nothing before 2016; from about 100 texts in 2019 to 956 in 2025; research grows by 60% a year. 2026 already exceeds 2025 by October. The turning point is 2020: Libra in 2019, COVID, and the ECB's report on a digital euro in October 2020.



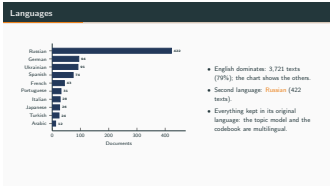
Languages



- English dominates: 3,721 texts (79%); the chart shows the others.
- Second language: Russian (422 texts).
- Everything kept in its original language: the topic model and the codebook are multilingual.

Slide 15 **Time: 0:20** clock at the end of the slide: **9:10** / 25:55

Say: 79% English, but Russian is a strong second (422 texts), mostly on the digital ruble. English-only databases would miss about a fifth of what economists write on CBDCs. Everything is kept in its language: the topic model and the codebook are multilingual. *If short of time: skip.*



Who writes

Research economists: outlets, most cited, laws

Main outlets

SSRN Electronic Journal	461
RePEc: Research Papers in Economics	164
Zenodo (CERN repository)	126
CyberLeninka (Russian repository)	70
Springer eBooks	68
Routledge eBooks	56
Ekonomika i Upravlenie	47

Bibliometric laws. Growth 60%/year (2017–2025). Lotka: $f(n) \propto n^{-3.3}$, 83% of authors publish once. Bradford: 21 core sources hold 1/3 of the papers (zones 21 : 310 : 1,148).

Most cited papers (global citations)

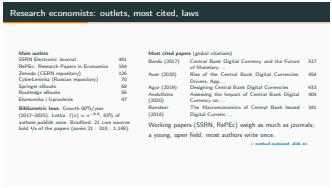
Bordo (2017)	Central Bank Digital Currency and the Future of Monetary...	517
Auer (2020)	Rise of the Central Bank Digital Currencies: Drivers, App...	454
Agur (2019)	Designing Central Bank Digital Currencies	433
Andolfatto (2020)	Assessing the Impact of Central Bank Digital Currency on...	404
Barrdear (2016)	The Macroeconomics of Central Bank Issued Digital Currenc...	341

Working papers (SSRN, RePEc) weigh as much as journals; a young, open field: most authors write once.

▷ method explained: slide 41

Slide 16 **Time: 0:40** clock at the end of the slide: **9:50** / 25:55

Say: The biggest outlets are working-paper repositories, SSRN and RePEc, not journals: the debate happens before peer review. The most cited papers are the classics from central bank and IMF economists: Bordo and Levin, Auer, Agur, Andolfatto, Barrdear and Kumhof. Lotka: 83% of authors write only once on CBDCs (exponent 3.3, against about 2 in mature fields). Bradford: 21 sources publish a third of the papers, the last third is spread over 1,148. A young, open field, around a small core.



Who speaks: central banks, institutions, research centres

Speeches, by central bank

European Central Bank	118
Deutsche Bundesbank	63
Bank of France	49
Hong Kong Monetary Authority	23
Federal Reserve Board	21
Bank of England	21
Bank of Italy	19
Bank of Japan	14

Documents, by institution

European Central Bank	98
Bank of Japan	27
Sveriges Riksbank	22
Deutsche Bundesbank	21
Swiss National Bank	9
Central Bank of The Bahamas	7
Bank of Jamaica	6
Bank of Canada	4

Research, by author institution

Financial University	63
European Central Bank	61
International Monetary Fund	39
Bank of Canada	38
The Russian Presidential Acad...	33
Bank for International Settle...	32
Moscow State Institute of Int...	29
Lomonosov Moscow State Univer...	29

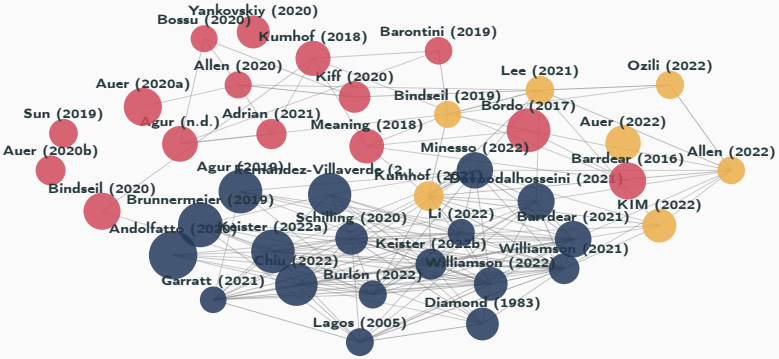
European Central Bank is the most vocal central bank: the digital euro makes it the hub of the debate.

Slide 17 **Time: 0:35** clock at the end of the slide: **10:25** / 25:55

Say: On the institutional side, the European Central Bank dominates: 118 speeches and 98 documents, because of the digital euro. Then the Bundesbank, the Banque de France, Hong Kong, the Fed, the Bank of England. In research, the ECB, the IMF, the Bank of Canada and the BIS are among the top institutions, next to Russian universities (Financial University, Moscow State).



Co-citation: the intellectual base



Most cited within the corpus

Andolfatto (2020)	264
Brunnermeier (2019)	218
Bordo (2017)	204
Agur (2019)	204
Keister (2022)	203
Fernández-Villaverde (2020)	199
Chiu (2022)	191
Auer (2020)	144

Clusters (largest members):

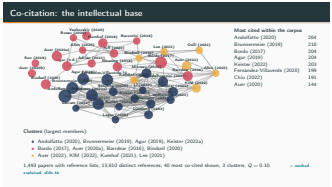
- Andolfatto (2020), Brunnermeier (2019), Agur (2019), Keister (2022a)
- Bordo (2017), Auer (2020a), Barrdear (2016), Bindseil (2020)
- Auer (2022), KIM (2022), Kumhof (2021), Lee (2021)

1,493 papers with reference lists, 13,810 distinct references; 40 most co-cited shown, 3 clusters, $Q = 0.10$. ▷ method

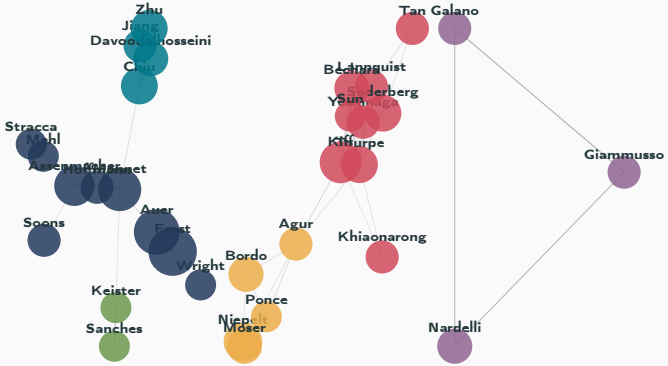
explained: slide 39

Slide 18 **Time: 0:50** clock at the end of the slide: **11:15** / 25:55

Say: Two references are linked when the same paper cites them both; links are weighted so that popular classics are not linked to everything. Three groups: Andolfatto, Brunnermeier, Agur, Keister (banking and design); Bordo, Auer, Barrdear, Bindseil (monetary policy and the first central bank studies); Auer 2022 and the recent Asian and design work. But modularity is only 0.10: the groups overlap. **There are no separate schools**: everyone builds on the same core. **If asked**: node area = citations; darker link = more often co-cited than expected.



Co-authorship: research communities



Most productive authors

Ulrich Bindseil	20
Peterson K Ozili	20
Jon Frost	16
Annelieke Mooij	16
Raphael Auer	14
Cyril Monnet	12
John Kiff	11
Katrin Assenmacher	10

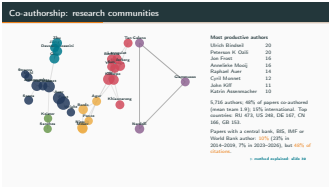
5,716 authors; 48% of papers co-authored (mean team 1.9); 15% international. Top countries: RU 473, US 248, DE 167, CN 166, GB 153.

Papers with a central bank, BIS, IMF or World Bank author: 10% (23% in 2014–2019, 7% in 2023–2026), but 48% of citations.

▷ method explained: slide 39

Slide 19 **Time: 0:45** clock at the end of the slide: **12:00** / 25:55

Say: Teams are small: half the papers have one author, a team averages 1.9 people, 15% are international. Each island on the map is a team: Bindseil's at the ECB, Auer and Frost at the BIS, Ozili alone but very productive. Key fact: papers with a central bank, BIS, IMF or World Bank author are 10% of papers but get **48% of citations**. Their share fell from 23% to 7%: new papers come from outside these institutions, but the field still cites the institutions.



Who says what: research economists vs central bankers

Words most over-used by each group compared with the other (z-scores):

Research economists (259,105 words)		**Central bankers** (232,446 words)	
legal	23	payments	28
analysis	21	euro	22
development	21	european	18
research	20	europe	18
monetary	20	area	18
impact	19	services	15
model	18	innovation	14
implementation	17	conference	14

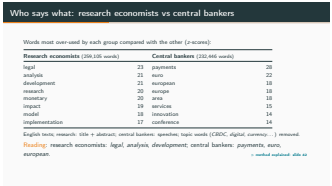
English texts; research: title + abstract; central bankers: speeches; topic words (*CBDC, digital, currency...*) removed.

Reading: research economists: *legal, analysis, development*; central bankers: *payments, euro, european*.

▷ method explained: slide 42

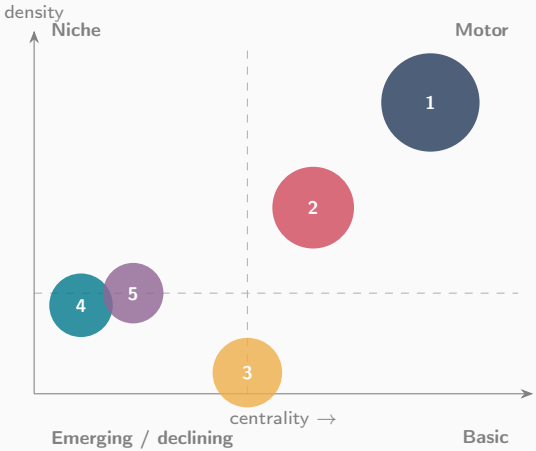
Slide 20 **Time: 0:40** clock at the end of the slide: **12:40** / 25:55

Say: Which words does each group use more than the other, once chance is accounted for? Research economists: legal, analysis, development, model, impact: they *analyse* CBDCs. Central bankers: payments, euro, Europe, services, innovation: they talk about *their* project, the digital euro, as a payment instrument. **If asked**: the method shrinks rare words, so a word used twice by one group cannot top the list.



What is written

Thematic map: which themes drive the field?



Keyword clusters placed by centrality (links to other themes) and density (internal cohesion).

	Theme (top keywords)	**Quadrant**
1	digital transformation in financial services, monetary policy, moneta. . .	motor
2	blockchain technology applications and security, digital transformati. . .	motor
3	fintech, crowdfunding, digital finance, blockchain, fintech	basic
4	global financial crisis and policies, international payments, stablec. . .	emerging or declining
5	financial inclusion, digital payments, india	niche

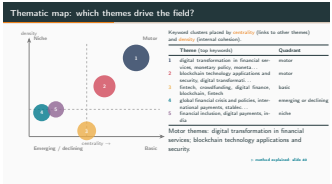
Motor themes: digital transformation in financial services; blockchain technology applications and security.

▷ method explained: slide 40

Slide 21 **Time: 0:45** clock at the end of the slide: **13:25** / 25:55

Say: Each bubble is a group of keywords that appear together. Horizontal: how linked the theme is to the rest of the field; vertical: how developed it is inside. Motor themes (top right): digital transformation of finance with monetary policy, and blockchain technology. Fintech is a basic theme. Financial inclusion with India and digital payments is a niche. Global financial crisis and stablecoins sits in emerging or declining.

If asked: quadrants are split at the medians, so positions are relative.



How the themes evolve

Keyword (% of the period's papers) *papers in period*	**2014–2019** *136*	**2020–2022** *755*	**2023–2026** *2,919*	**Change**
financial inclusion	5.9	8.2	15.8	+7.6
digital euro	2.9	7.8	11.4	+3.6
global financial crisis and policies	2.2	8.7	11.8	+3.0
financial stability	24.3	12.6	15.6	+3.0
fintech, crowdfunding, digital finance	48.5	48.7	49.8	+1.0
digital ruble	0.0	13.9	14.8	+0.9
european monetary and fiscal policies	16.9	10.3	10.3	−0.1
digital transformation in financial ser. . .	72.1	74.7	74.2	−0.5
digital transformation in law	10.3	17.9	17.3	−0.6
banking stability, regulation, efficien. . .	22.1	16.4	15.5	−0.9
monetary policy and economic impact	51.5	23.0	20.7	−2.4
digital payments	8.8	12.6	10.2	−2.4

Sorted by change between the last two periods. Darker = more frequent.

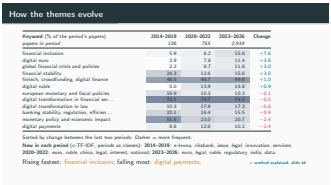
New in each period (c-TF-IDF, periods as classes): **2014–2019**: *e-krona, riksbank, issue, legal, innovation, services*; **2020–2022**: *euro, ruble, china, legal, interest, national*; **2023–2026**: *euro, legal, ruble, regulatory, india, data*.

Rising fastest: financial inclusion; falling most: digital payments.

▷ method explained: slide 45

Slide 22 **Time: 0:35** clock at the end of the slide: **14:00** / 25:55

Say: Share of each period's papers carrying each keyword. Financial inclusion rises fastest, from 6% to 16%; the digital euro and the digital ruble appear; monetary policy falls from 52% to 21%. The vocabulary specific to each period goes from the Swedish e-krona, to the euro, ruble and China, to India and regulation.



Topics found by meaning (SBERT + BERTopic)

	Topic (top c-TF-IDF terms)	**Texts**	**% of the period's texts**			**Adjusted, points**	
			2014–2019	**2020–2022**	**2023–2026**	**change**	**CB**
1	cryptocurrencies, implementation, private, tr...	470	18.5	13.5	8.5	**-13.2**	**-8.6**
2	design, retail, legal, private	439	10.4	11.2	8.6	-4.0	**-9.4**
3	uncertainty, rates, design, rate	349	14.2	8.7	6.7	**-9.6**	**-5.9**
4	dollar, trade, debt, reserve	326	5.7	5.8	7.5	+0.8	**-5.0**
5	deposits, interest, deposit, rate	319	2.8	5.0	7.3	+3.6	**-7.7**
6	governor, innovation, fintech, kong	306	17.5	11.2	4.7	-1.2	**+35.7**
7	russian, russia, ruble, federation	238	0.0	5.1	4.9	**+4.7**	**-5.2**
8	china, rmb, e-cny, yuan	213	1.4	5.3	4.6	+2.7	**-5.2**
9	cashless, evolution, riksbank, forms	204	6.6	4.9	4.1	**-3.8**	**-1.8**
10	fintech, technologies, services, innovation	199	7.1	2.9	4.4	**-3.0**	-0.5
11	euro, european, ecb, area	186	0.5	3.6	4.2	**+3.7**	**+5.6**
12	cryptocurrencies, crypto, bitcoin, cryptocurr...	186	9.0	4.0	3.6	**-5.8**	-1.1

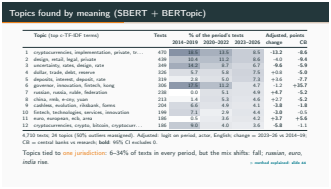
4,710 texts; 24 topics (50% outliers reassigned). Adjusted: logit on period, actor, English; change = 2023–26 vs 2014–19; CB = central banks vs research; **bold**: 95% CI excludes 0.

Topics tied to one jurisdiction: 6–34% of texts in every period, but the mix shifts: fall; *russian*, *euro*, *india* rise.

▷ method explained: slide 44

Slide 23 **Time: 1:00** clock at the end of the slide: **15:00** / 25:55

Say: Here texts are grouped by meaning, not by keywords, so a Russian and an English paper on the digital ruble land in the same topic. 24 topics; the 12 largest are shown. Falling: cryptocurrencies and private money (−13 points), uncertainty and rates (−10), crypto and bitcoin (−6). Rising: Russia and the ruble (+5), the euro and the ECB (+4). Topics tied to one jurisdiction (China, Russia, euro, India) are a stable share, but the mix shifts: China and Sweden fall, Russia, the euro and India rise. The literature follows national projects. The "adjusted" columns compare like with like: same type of author and language.



Topics found by meaning (SBERT + BERTopic)

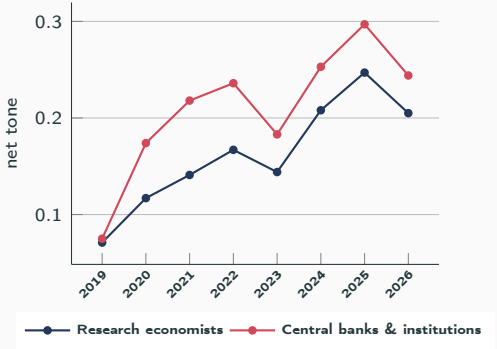
	Topic (top c-TF-IDF terms)	Texts	% of the period's texts			Adjusted, points	
			2014–2019	2020–2022	2023–2026	change	CI
1	cryptocurrencies, implementation, private, tr…	470	18.5	13.5	8.5	-13.2	-8.6
2	design, retail, legal, private	438	10.4	11.2	8.6	-4.0	-6.4
3	uncertainty, rates, design, rate	348	14.2	8.7	6.7	-9.6	-5.9
4	dollar, trade, debt, reserve	326	5.7	5.8	7.5	+0.8	-5.0
5	deposits, interest, deposit, rate	319	2.3	5.0	7.3	+3.5	-3.7
6	governor, innovation, fintech, kong	306	0.0	11.2	4.7	-1.2	+35.7
7	russian, russia, ruble, federation	228	0.0	5.1	4.9	+4.7	-5.2
8	bitcoin, crypto, assets, digital	211	0.0	4.8	4.1	+3.8	-4.8
9	cashless, evolution, vitukaitė, forms	204	0.0	4.0	4.1	-3.0	-1.0
10	fintech, technologies, services, innovation	190	7.1	2.9	4.4	-3.0	-0.5
11	euro, european, ecb, area	186	1.1	3.0	6.2	+3.7	+5.6
12	cryptocurrencies, crypto, bitcoin, cryptocurr…	186	9.0	4.0	3.6	-5.8	-1.1

6,730 texts; 24 topics (92% outliers reassigned). Adjusted: logit on period, actor, English; change = 2023–26 vs 2014–19; CI = central banks vs research; bold: 95% CI excludes 0.

Topics tied to one jurisdiction: 6–34% of texts in every period, but the mix shifts: fall: *russian, euro, india* rise.

How: tone and stance

Tone: how positive is what economists say about CBDCs?



	Texts	Sent.	%+	%−	Net tone
Research economists	2,306	7,519	21	5	0.19 ± 0.01
Central banks & institutions	630	7,576	18	3	0.22 ± 0.01

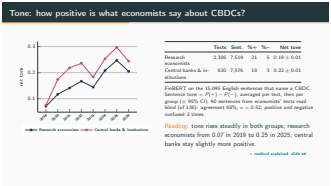
FinBERT on the 15,095 English sentences that name a CBDC. Sentence tone $= P(+) - P(-)$, averaged per text, then per group ($\pm$ 95% CI). 90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

Reading: tone rises steadily in both groups; research economists from 0.07 in 2019 to 0.25 in 2025; central banks stay slightly more positive.

▷ method explained: slide 48

Slide 24 **Time: 1:00** clock at the end of the slide: **16:00** / 25:55

Say: Tone is how positive the language is. We keep only the sentences that name a CBDC and score them with FinBERT, a model trained on financial text: tone = probability positive minus probability negative, averaged per text, so that every text counts the same. Result: tone rises steadily in both groups. Research economists go from 0.07 in 2019 to 0.25 in 2025; central banks are slightly more positive throughout (0.22 against 0.19 on average). Checked by reading 90 sentences: kappa 0.52, and positive and negative are confused only 3 times. **If asked "why only English?"**: FinBERT is English; the other languages are covered by stance and motivations.



Do policy events move the tone?

Date	Event	Before	After	Δtone	p
Jun 2019	Facebook announces Libra	23	42	-0.041	0.33
Oct 2020	ECB digital euro report	46	113	+0.003	0.89
Oct 2020	Bahamas launches Sand Dollar	57	111	-0.020	0.55
Jul 2021	ECB investigation phase	110	102	+0.020	0.56
Oct 2021	Nigeria launches eNaira	115	103	+0.050	0.24
Jan 2022	Fed CBDC discussion paper	102	122	+0.022	0.51
Mar 2022	US order on digital assets	108	129	+0.004	0.87
Nov 2022	India e-rupee pilot	130	128	+0.011	0.71
Feb 2023	UK digital pound consultation	132	148	-0.008	0.77
Jun 2023	EU digital euro bill	145	211	-0.043	0.30
Oct 2023	ECB preparation phase	169	217	-0.025	0.48
Jan 2025	US order banning a US CBDC	193	230	+0.018	0.60

For each event: mean tone of the texts in the 180 days *after* minus the 180 days *before*. Compared with 500 random dates: p = share of random dates with a larger $|\Delta|$ (95th percentile: 0.084).

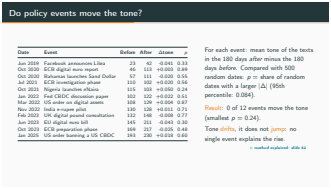
Result: 0 of 12 events move the tone (smallest $p = 0.24$).

Tone drifts, it does not jump: no single event explains the rise.

▷ method explained: slide 52

Slide 25 **Time: 0:50** clock at the end of the slide: **16:50** / 25:55

Say: Could one big event explain the rise in tone? For each of 12 events, we compare the tone 6 months after with 6 months before, and we compare that change with the same change around 500 random dates. Result: **none of the 12 events moves the tone**; the smallest p-value is 0.24. Libra: -0.04; eNaira: $+0.05$; ordinary dates move that much. So tone *drifts*, it does not *jump*: the change is gradual.



Stance: for or against CBDCs?

Level, read by hand, % [95% CI]

	n	Favourable	Critical
Research economists	50	8 **[3–19]**	2 **[0–10]**
Central banks & institutions	50	52 **[39–65]**	6 **[2–16]**

By reading: 52% of central bank texts are favourable; research economists are mostly neutral (90%): they analyse, central banks advocate.

Change (local LLM, 575 texts sampled per actor × period), % favourable

	2014–2019	2020–2022	2023–2026	Change, pts
Research economists	26	38	40	**+14** **[+1, +27]**
Central banks & institutions	25	54	65	**+40** **[+26, +53]**

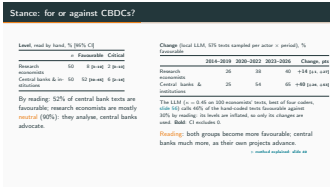
The LLM ($\kappa = 0.45$ on 100 economists' texts, best of four coders, slide 56) calls 46% of the hand-coded texts favourable against 30% by reading: its levels are inflated, so only its *changes* are used. **Bold**: CI excludes 0.

Reading: both groups become more favourable; central banks much more, as their own projects advance.

▷ method explained: slide 49

Slide 26 **Time: 1:00** clock at the end of the slide: **17:50** / 25:55

Say: Stance is different from tone: is the author for or against issuing a CBDC? Levels, by reading 50 texts per group: research economists are 90% neutral, only 8% favourable; central banks are 52% favourable. **Researchers analyse, central banks advocate.** Change, with a local language model on a random sample: both groups become more favourable between 2014–19 and 2023–26: research +14 points, central banks +40 points. The model is the best of four coders but over-calls "favourable", so we use it only for changes, where a constant bias cancels; levels come from reading.



Why: stated motivations

Coding stated motivations

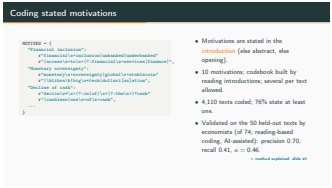
```
MOTIVES = {
  "Financial inclusion":
      r"financial\s+inclusion|unbanked|underbanked"
      r"|access\s+to\s+(?:financial\s+services|finance)",
  "Monetary sovereignty":
      r"monetary\s+sovereignty|global\s+stablecoin"
      r"|\blibra\b|big\s*tech|dollari[sz]ation",
  "Decline of cash":
      r"declin\w*\s+(?:in|of)\s+(?:the\s+)?cash"
      r"|cashless|use\s+of\s+cash",
  ...
}
```

- Motivations are stated in the introduction (else abstract, else opening).
- 10 motivations; codebook built by reading introductions; several per text allowed.
- 4,110 texts coded; 76% state at least one.
- Validated on the 50 held-out texts by economists (of 74; reading-based coding, AI-assisted): precision 0.70, recall 0.41, $\kappa = 0.46$.

▷ method explained: slide 47

Slide 27 **Time: 0:35** clock at the end of the slide: **18:25** / 25:55

Say: Why, in their own words. Motivations are stated in the introduction, so that is where we look. Ten motivations, written as phrase lists in seven languages after reading 100 introductions, for example “unbanked” or “access to financial services” for financial inclusion. Checked on 50 other texts by economists: precision 0.70, recall 0.41, kappa 0.46.



Why: what economists say CBDCs are for

Motivation	% of texts 2014–2019	% of texts 2020–2022	% of texts 2023–2026	Adjusted effect, points [95% CI] Change 2023–26 vs 2014–19	Adjusted effect, points [95% CI] Central banks vs research
Financial stability	30.8	26.3	29.3	-1.6 [-8, +5]	**-7.7 [-11, -4]**
Monetary policy	36.8	27.4	24.4	**-15.1 [-22, -8]**	**-13.6 [-16, -11]**
Technology (DLT, programmability)	22.5	21.6	25.8	+1.6 [-5, +8]	**-9.1 [-12, -6]**
Cross-border payments	13.2	19.0	22.5	**+8.7 [+3, +13]**	**-7.8 [-11, -5]**
Payment efficiency	10.4	12.1	17.4	**+6.6 [+1, +11]**	**-3.1 [-6, -0]**
Monetary sovereignty	15.9	16.8	13.6	-2.0 [-8, +3]	-2.4 [-5, +0]
Financial inclusion	3.8	8.9	16.8	**+12.3 [+8, +15]**	**-9.5 [-12, -7]**
Geopolitics and sanctions	2.7	7.2	15.0	**+12.4 [+9, +15]**	+0.3 [-2, +3]
Privacy and surveillance	7.1	10.5	13.6	**+6.4 [+2, +10]**	**-2.8 [-5, -0]**
Decline of cash	14.8	9.0	7.4	**-6.5 [-12, -2]**	+1.2 [-1, +4]

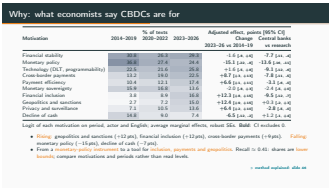
Logit of each motivation on period, actor and English; average marginal effects, robust SEs. **Bold**: CI excludes 0.

- Rising: geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts). Falling: monetary policy (−15 pts), decline of cash (−7 pts).
- From a monetary-policy instrument to a tool for inclusion, payments and geopolitics. Recall ≈ 0.41: shares are lower bounds; compare motivations and periods rather than read levels.

▷ method explained: slide 46

Slide 28 **Time: 1:00** clock at the end of the slide: **19:25** / 25:55

Say: Rising: geopolitics and sanctions (+12 points), financial inclusion (+12), cross-border payments (+9), payment efficiency and privacy (+6). Falling: monetary policy (−15), the decline of cash (−7). **CBDCs move from a monetary-policy instrument to a tool for inclusion, payments and geopolitics.** Changes are adjusted for who writes and in which language. Because recall is about 0.4, read the changes, not the levels.



Is the codebook the best coder?

Coder	P	R	F_1	κ	κ, all
Codebook (regex)	0.70	0.41	0.52	0.46	0.50
Codebook OR classifier	0.42	0.60	0.50	0.39	0.42
SBERT, classifier (100 labels)	0.36	0.37	0.37	0.25	0.28
Local LLM, Qwen2.5 7B	0.30	0.61	0.40	0.25	0.28
Local LLM, Qwen2.5 3B	0.31	0.23	0.26	0.15	0.16
SBERT, definitions (zero-shot)	0.21	0.71	0.32	0.12	0.15
NLI zero-shot (mDeBERTa)	0.22	0.39	0.28	0.11	0.14

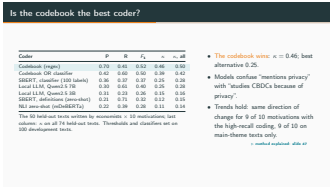
The 50 held-out texts written by economists $\times$ 10 motivations; last column: κ on all 74 held-out texts. Thresholds and classifiers set on 100 development texts.

- The codebook wins: $\kappa = 0.46$; best alternative 0.25.
- Models confuse "mentions privacy" with "studies CBDCs because of privacy".
- Trends hold: same direction of change for 9 of 10 motivations with the high-recall coding, 9 of 10 on main-theme texts only.

▷ method explained: slide 47

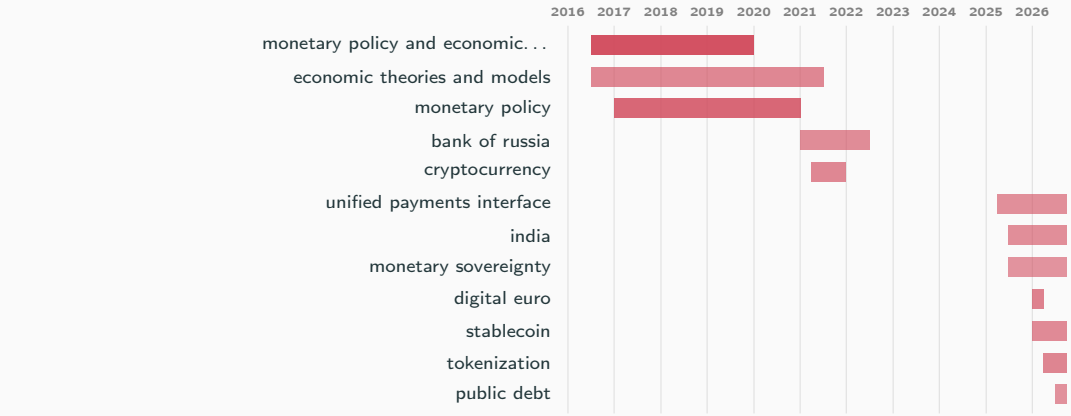
Slide 29 **Time: 0:35** clock at the end of the slide: **20:00** / 25:55

Say: We tested six machine coders against our reading. The codebook wins: kappa 0.46, the best alternative 0.39 and the language models 0.25. Models confuse "mentions privacy" with "studies CBDCs *because of* privacy". The trends hold with a high-recall coding: 9 of 10 motivations move the same way. *If short of time: one sentence.*



When: the environment

Which themes burst, and when

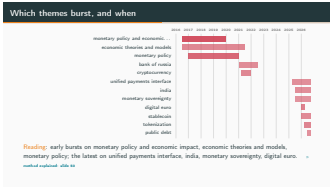


Reading: early bursts on monetary policy and economic impact, economic theories and models, monetary policy; the latest on unified payments interface, india, monetary sovereignty, digital euro. ▷

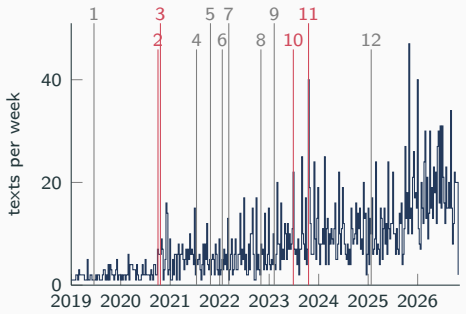
method explained: slide 50

Slide 30 **Time: 0:40** clock at the end of the slide: **20:40** / 25:55

Say: Kleinberg's algorithm finds when a keyword suddenly becomes frequent, with a cost for entering a burst so that noise is ignored. Early bursts: monetary policy, economic impact, theory and models. Latest: India and its payment system (UPI), monetary sovereignty, the digital euro. From theory to national projects.



Attention around policy events



	Date	Event	Ratio	z	p
1	Jun 2019	Facebook announces Libra	0.3	-1.3	0.83
2	Oct 2020	ECB digital euro report	2.9	+4.0	0.01
3	Oct 2020	Bahamas launches Sand Dollar	3.0	+4.7	0.01
4	Jul 2021	ECB investigation phase	0.5	-1.9	0.93
5	Oct 2021	Nigeria launches eNaira	0.8	-0.7	0.69
6	Jan 2022	Fed CBDC discussion paper	1.4	+1.2	0.24
7	Mar 2022	US order on digital assets	0.6	-1.3	0.84
8	Nov 2022	India e-rupee pilot	1.0	-0.2	0.55
9	Feb 2023	UK digital pound consultation	0.7	-1.0	0.76
10	Jun 2023	EU digital euro bill	1.7	+2.9	0.05
11	Oct 2023	ECB preparation phase	3.3	+10.0	0.00
12	Jan 2025	US order banning a US CBDC	1.2	+0.9	0.29

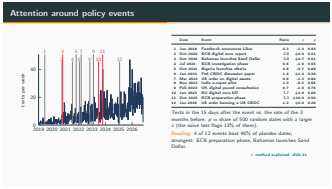
Texts in the 15 days after the event vs. the rate of the 3 months before. p = share of 500 random dates with a larger z (the naive test flags 13% of them).

Reading: 4 of 12 events beat 95% of placebo dates; strongest: ECB preparation phase, Bahamas launches Sand Dollar.

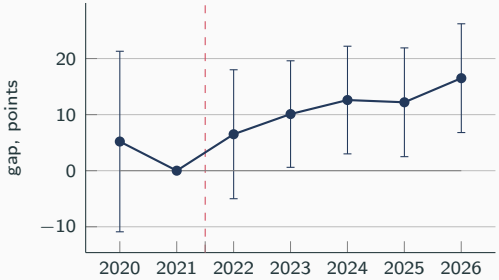
▷ method explained: slide 51

Slide 31 **Time: 0:50** clock at the end of the slide: **21:30** / 25:55

Say: Does writing on CBDCs jump after a policy event? We compare the 15 days after each event with the rate of the 3 months before, and with 500 random dates, because attention grows and comes in waves. **4 of 12 events beat 95% of random dates**: the ECB report and the Sand Dollar launch (October 2020), the EU digital euro bill (June 2023), and the ECB preparation phase (October 2023, three times the usual rate). Libra, the eNaira, the US orders do not. Red lines on the chart are the significant events.



The 2022 sanctions: did geopolitics enter the discussion?



Gap in the share stating *geopolitics and sanctions*: texts about Russia (or in Russian) minus the others, relative to 2021; 95% CI.

756 texts about Russia among 3,772 (2020–2026). Pooled effect after 2021, every motivation:

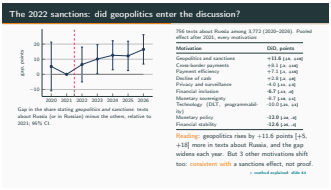
Motivation	DiD, points
Geopolitics and sanctions	**+11.6** [+5, +18]
Cross-border payments	+8.1 [-2, +18]
Payment efficiency	+7.1 [-1, +15]
Decline of cash	+2.8 [-2, +8]
Privacy and surveillance	-4.0 [-11, +3]
Financial inclusion	**-6.7** [-13, -0]
Monetary sovereignty	-8.7 [-18, +1]
Technology (DLT, programmability)	-10.0 [-21, +1]
Monetary policy	**-12.0** [-24, -0]
Financial stability	**-12.6** [-25, -1]

Reading: geopolitics rises by +11.6 points [+5, +18] more in texts about Russia, and the gap widens each year. But 3 other motivations shift too: consistent with a sanctions effect, not proof.

▷ method explained: slide 53

Slide 32 **Time: 1:00** clock at the end of the slide: **22:30** / 25:55

Say: After the 2022 sanctions, did geopolitics enter the discussion? We compare texts about Russia with the others, before and after 2021. Geopolitics and sanctions rises by **11.6 points more** in texts about Russia [+5, +18], and the gap widens every year. But other motivations move too (monetary policy, financial stability, inclusion), and we have only one clean year before (2020). So: **consistent with** a sanctions effect, not proof.



The discussion and its environment

Putting it together, period by period

	2014–2019	2020–2022	2023–2026
Policy events (bold: attention reacts)	**Facebook announces Libra**	ECB digital euro report; Bahamas launches Sand Dollar; **ECB investigation phase**; **Nigeria launches eNaira**; **Fed CBDC discussion paper**; **US order on digital assets**; India e-rupee pilot	**UK digital pound consultation**; EU digital euro bill; ECB preparation phase; **US order banning a US CBDC**
Texts per year	**53**	**324**	**840**
Bursting keywords	**monetary policy and economic impact, economic theories and models, monetary policy**	**bank of russia, cryptocurrency**	**unified payments interface, india, monetary sovereignty**
Most frequent motivations	**monetary policy, financial stability, technology**	**monetary policy, financial stability, technology**	**financial stability, technology, monetary policy**
Net tone (research / central banks)	**0.06 / 0.10**	**0.15 / 0.22**	**0.21 / 0.24**
Favourable, LLM (research / central banks)	**26% / 25%**	**38% / 54%**	**40% / 65%**

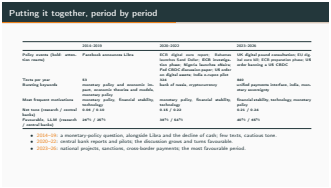
- 2014–19: a monetary-policy question, alongside Libra and the decline of cash; few texts, cautious tone.
- 2020–22: central bank reports and pilots; the discussion grows and turns favourable.
- 2023–26: national projects, sanctions, cross-border payments; the most favourable period.

Slide 33 **Time: 1:00** clock at the end of the slide: **23:30** / 25:55

Say: Three periods.

- 2014–19: a monetary-policy question, alongside Libra and the decline of cash; few texts, cautious tone.
- 2020–22: central bank reports and pilots; the discussion grows and turns favourable.
- 2023–26: national projects, sanctions, cross-border payments; the most favourable period.

The discussion moves with its environment: more texts, more favourable, new motivations, as projects advance.



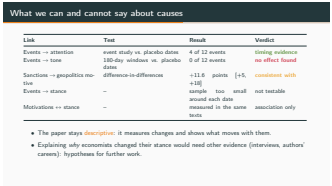
What we can and cannot say about causes

Link	Test	Result	Verdict
Events → attention	event study vs. placebo dates	4 of 12 events	**timing evidence**
Events → tone	180-day windows vs. placebo dates	0 of 12 events	**no effect found**
Sanctions → geopolitics motive	difference-in-differences	+11.6 points [+5, +18]	**consistent with**
Events → stance	–	sample too small around each date	not testable
Motivations ↔ stance	–	measured in the same texts	association only

- The paper stays descriptive: it measures changes and shows what moves with them.
- Explaining *why* economists changed their stance would need other evidence (interviews, authors' careers): hypotheses for further work.

Slide 34 **Time: 0:50** clock at the end of the slide: **24:20** / 25:55

Say: What can we say about causes? Events move attention (4 of 12): timing evidence. Events do not move tone (0 of 12). The sanctions are consistent with the rise of geopolitics, not proof. Events and stance: samples too small around each date. Motivations and stance are measured in the same texts: association only. So the paper stays descriptive; explaining *why* economists changed their stance would need other evidence, interviews or authors' careers.



Robustness

Robustness

Specification	Texts	Why: change, points inclusion	Why: change, points monetary policy	How: tone CB − research	How: stance fav. CB − research	When events
Baseline: economists	4,710	+12.3 [+8, +15]	−15.1 [-22, -8]	+0.03	+16	4/12
Main theme only	3,513	+14.9 [+10, +18]	−16.0 [-25, -7]	+0.01	+13	4/12
English only	3,721	+13.7 [+9, +16]	−16.0 [-23, -7]	+0.03	+16	4/12
Journal articles only (research)	3,171	+12.3 [+6, +15]	−16.8 [-25, -8]	+0.01	+11	4/12
Without 2026 (partial year)	3,547	+11.8 [+7, +15]	−15.3 [-23, -8]	+0.03	+15	4/12
High-recall motivation coding	4,710	+15.6 [+10, +19]	−8.8 [-16, -2]	+0.03	+16	4/12
All disciplines (not only economists)	8,213	+13.2 [+10, +15]	−15.0 [-21, -9]	+0.02	+15	4/12

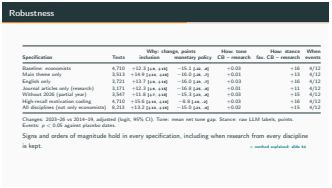
Changes: 2023–26 vs 2014–19, adjusted (logit, 95% CI). Tone: mean net tone gap. Stance: raw LLM labels, points. Events: $p < 0.05$ against placebo dates.

Signs and orders of magnitude hold in every specification, including when research from every discipline is kept.

▷ method explained: slide 54

Slide 35 **Time: 0:45** clock at the end of the slide: **25:05** / 25:55

Say: We re-ran the headline results seven ways: main theme only, English only, journal articles only, without the partial year 2026, with a high-recall motivation coding, and with research from every discipline. Signs and orders of magnitude hold everywhere: inclusion $+12$ to $+16$ points, monetary policy -9 to -17, the same 4 events.



Conclusion

Takeaways and limits

How economists' discussion of CBDCs evolved

- Who: 4,710 texts; central bank economists write 10% of the papers but get 48% of citations.
- What: from monetary policy and design to inclusion and national projects (digital ruble, rupee, e-CNY).
- How: tone and stance become more favourable; research economists stay mostly neutral, central banks advocate.
- Why: rising geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts); falling monetary policy (−15 pts), decline of cash (−7 pts).
- When: 4 of 12 events move attention, 0 of 12 move tone; geopolitics rises after the 2022 sanctions.

Limits

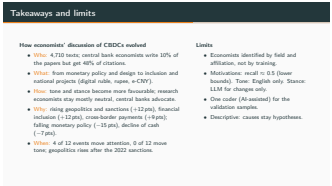
- Economists identified by field and affiliation, not by training.
- Motivations: recall ≈ 0.5 (lower bounds). Tone: English only. Stance: LLM for changes only.
- One coder (AI-assisted) for the validation samples.
- Descriptive: causes stay hypotheses.

Slide 36 **Time: 0:50** clock at the end of the slide: **25:55** / 25:55

Say: To conclude:

- Who: 4,710 texts; central bank economists write 10% of papers but get half the citations.
- What: from monetary policy and design to inclusion and national projects.
- How: tone and stance become more favourable; researchers stay neutral, central banks advocate.
- Why: geopolitics, inclusion and cross-border payments rise; monetary policy and cash fall.
- When: events move attention, not tone; geopolitics rises after the 2022 sanctions.

Limits: economists identified by field and employer, not training; motivations are lower bounds; tone in English only; descriptive. **Thank you; we are happy to take questions.**



Methods explained

Methods explained: how to read this appendix

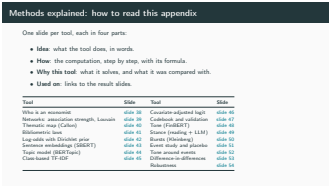
One slide per tool, each in four parts:

- **Idea**: what the tool does, in words.
- **How**: the computation, step by step, with its formula.
- **Why this tool**: what it solves, and what it was compared with.
- **Used on**: links to the result slides.

Tool	Slide	Tool	Slide
Who is an economist	slide 38	Covariate-adjusted logit	slide 46
Networks: association strength, Louvain	slide 39	Codebook and validation	slide 47
Thematic map (Callon)	slide 40	Tone (FinBERT)	slide 48
Bibliometric laws	slide 41	Stance (reading + LLM)	slide 49
Log-odds with Dirichlet prior	slide 42	Bursts (Kleinberg)	slide 50
Sentence embeddings (SBERT)	slide 43	Event study and placebo	slide 51
Topic model (BERTopic)	slide 44	Tone around events	slide 52
Class-based TF-IDF	slide 45	Difference-in-differences	slide 53
		Robustness	slide 54

Slide 37

Appendix: shown only if asked (0:00 in the budget). Use this index to jump to the method a question is about.



Method: who counts as an economist

Idea

The course asks how *economists* discuss CBDCs, so each text must be traced to an economist or an economic institution.

How

- **Research**: kept if OpenAlex classifies it in *Economics, Econometrics and Finance* (primary topic, from title, abstract, venue, citations), or an author works at a central bank, the IMF or the BIS; without OpenAlex record: an economics venue.
- **Institutions**: central bank speeches; documents of central banks and international financial institutions.
- **Dropped**: other fields (computer science, law…), governments and parliaments, the press.

Why this tool

Authors' training is not recorded anywhere at scale; the field of the paper and the employer of the author are, and they are what defines a discipline in bibliometrics.

Speeches are kept as *institutional* voices: written with the bank's economics staff, even when the speaker is not an economist.

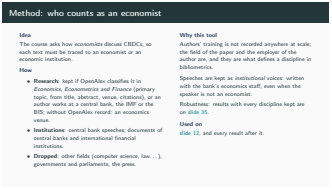
Robustness: results with every discipline kept are on slide 35.

Used on

slide 12, and every result after it.

Slide 38

Appendix: shown only if asked (0:00 in the budget). Key line: training is not recorded at scale; field and employer are what bibliometrics uses. Speeches = institutional voice. With every discipline kept, results hold (robustness).



Method: who counts as an economist

Idea

The course asks how economists discuss CBDCs, so each text must be traced to an economist or an economic institution.

How

- **Research**: kept if OpenAlex classifies it in *Economics, Econometrics and Finance* (primary topic, from title, abstract, venue, citations), or an author works at a central bank, the IMF or the BIS; without OpenAlex record: an economics venue.
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Speeches are kept as institutional voices: written with the bank's economics staff, even when the speaker is not an economist.

Robustness: results with every discipline kept are on slide 35.

Used on

slide 12, and every result after it.

Method: networks (association strength, Louvain)

Idea

Two items (references, authors, keywords) are linked when they appear in the same paper; groups of tightly linked items are schools, teams or themes.

How

1. Count c_{ij} = papers where i and j appear together; c_i = papers with i.
2. Normalise by association strength $s_{ij} = c_{ij}/(c_i c_j)$: the ratio of observed to expected co-occurrence.
3. Cluster with Louvain: move nodes between groups while the modularity $Q = \frac{1}{2m}\sum_{ij}\left[s_{ij} - \frac{k_i k_j}{2m}\right]\delta(g_i, g_j)$ rises.
4. Draw: force-directed layout; size $\propto \sqrt{c_i}$.

Why this tool

Raw counts favour popular items: two classics are co-cited often by chance. Association strength removes this (van Eck & Waltman, 2009); it is VOSviewer's default.

Louvain is fast and needs no number of clusters; Q tells how clear the grouping is.

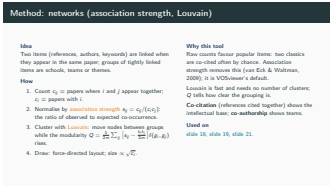
Co-citation (references cited together) shows the intellectual base; **co-authorship** shows teams.

Used on

slide 18, slide 19, slide 21.

Slide 39

Appendix: shown only if asked (0:00 in the budget). Why association strength: raw counts link popular items by chance. Why Louvain: fast, no number of clusters to choose; Q says how clear the grouping is (0.10: weak).



Method: thematic map (Callon)

Idea

Place each cluster of keywords on two axes: how much it is linked to the rest of the field, and how developed it is inside.

How

With the equivalence index $e_{ij} = c_{ij}^2/(c_i c_j)$:

$$\text{centrality} = 10 \sum_{i \in C,\, j \notin C} e_{ij}, \qquad \text{density} = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}$$

Quadrants, split at the medians:

- **motor**: central and dense; **basic**: central, loose;
- **niche**: dense, isolated; **emerging or declining**: neither.

Why this tool

A keyword list says what is frequent; the map says what *structures* the field. It is the standard tool of science mapping (Callon et al. 1991; Cobo et al. 2011; bibliometrix).

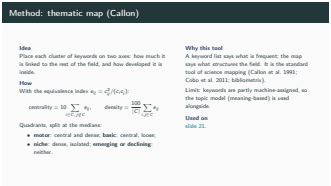
Limit: keywords are partly machine-assigned, so the topic model (meaning-based) is used alongside.

Used on

slide 21.

Slide 40

Appendix: shown only if asked (0:00 in the budget). Motor = central and dense; basic = central, loose; niche = dense, isolated; emerging or declining = neither. Limit: keywords are partly machine-assigned, hence the topic model alongside.



Method: bibliometric laws

Idea

Three regularities describe the shape of any scientific field: how fast it grows, how output is spread among authors, and among journals.

How

- **Growth**: compound annual rate $g = (N_{y_1}/N_{y_0})^{1/(y_1 - y_0)} - 1$.
- **Lotka**: share of authors with n papers $f(n) \propto n^{-\beta}$; β fitted on the log-log counts ($\beta \approx 2$ in mature fields).
- **Bradford**: sort sources by output, cut in three zones holding $1/3$ of papers each; the first zone is the core.

Why this tool

They situate the field: a high growth rate and a large β (most authors write once) mean a young, open field around a small core of outlets.

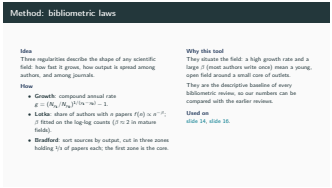
They are the descriptive baseline of every bibliometric review, so our numbers can be compared with the earlier reviews.

Used on

slide 14, slide 16.

Slide 41

Appendix: shown only if asked (0:00 in the budget). Growth 60%/year; Lotka exponent 3.3 (most authors write once); Bradford zones 21 : 310 : 1,148 (dispersed field).



Method: distinctive words (log-odds with Dirichlet prior)

Idea

Which words does one group use *more* than the other, once chance is accounted for?

How

For word w, group i vs. the other group j, with counts y, totals n, and a prior $\alpha_w = \alpha_0 \times$ (corpus frequency of w):

$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}$$

$$z_w = \delta_w \Big/ \sqrt{\frac{1}{y_{iw}+\alpha_w} + \frac{1}{y_{jw}+\alpha_w}}$$

Words are ranked by z_w.

Why this tool

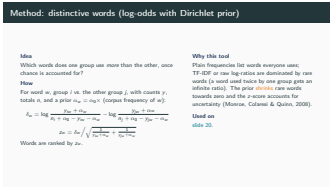
Plain frequencies list words everyone uses; TF-IDF or raw log-ratios are dominated by rare words (a word used twice by one group gets an infinite ratio). The prior shrinks rare words towards zero and the z-score accounts for uncertainty (Monroe, Colaresi & Quinn, 2008).

Used on

slide 20.

Slide 42

Appendix: shown only if asked (0:00 in the budget). The prior adds pseudo-counts proportional to each word's overall frequency: rare words are pulled to zero; the z-score accounts for uncertainty.



Method: sentence embeddings (SBERT)

Idea

Turn each text into a vector of 384 numbers such that texts with similar *meaning* get close vectors, whatever their words or language.

How

- A transformer network reads title + abstract (or opening) and averages its word representations into one vector $\mathbf{e}_t$.
- Model: `paraphrase-multilingual-MiniLM-L12-v2`, trained so that translations and paraphrases land together (50+ languages).
- Similarity: cosine $\cos(\mathbf{e}_s, \mathbf{e}_t) = \frac{\mathbf{e}_s \cdot \mathbf{e}_t}{\|\mathbf{e}_s\| \|\mathbf{e}_t\|}$.

Why this tool

Keywords and word counts treat "digital ruble" and its Russian original (*tsifrovoy rubl'*) as unrelated; embeddings do not. With 79% English and many Russian texts, this is what makes a single topic model possible.

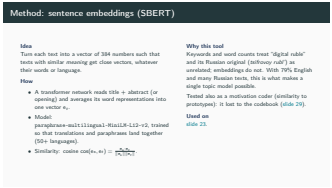
Tested also as a motivation coder (similarity to prototypes): it lost to the codebook (slide 29).

Used on

slide 23.

Slide 43

Appendix: shown only if asked (0:00 in the budget). Why: keywords treat "digital ruble" and *tsifrovoy rubl'* as unrelated; embeddings do not. Tested as a motivation coder too: lost to the codebook.



Method: topic model (BERTopic)

Idea

Find groups of texts about the same subject, without fixing the subjects in advance.

How

1. **Embed** each text (SBERT).
2. **Reduce** 384 dimensions to 5 with UMAP, which keeps neighbours together.
3. **Cluster** with HDBSCAN: dense regions become topics (at least 40 texts; leaf selection, which favours fine topics over one large blob); texts in no dense region are outliers, then assigned to the nearest topic.
4. **Label** each topic by its c-TF-IDF terms (slide 45).

Why this tool

LDA and STM count words, one language at a time, and need the number of topics. BERTopic works on meaning, across languages, and finds the number itself (Grootendorst, 2022).

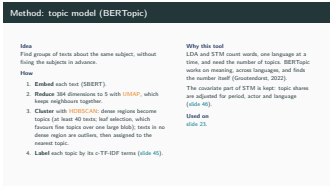
The covariate part of STM is kept: topic shares are adjusted for period, actor and language (slide 46).

Used on

slide 23.

Slide 44

Appendix: shown only if asked (0:00 in the budget). Why leaf selection: the default merged half the corpus into one topic. Why 40: smallest topic still large enough to compute shares by period.



Method: class-based TF-IDF (c-TF-IDF)

Idea

Which words characterise a class of texts (a topic, or a period) compared with all the others?

How

Merge the texts of each class c into one document; for term t:

$$W_{t,c} = \frac{\mathrm{tf}_{t,c}}{|c|} \log\left(1 + \frac{A}{f_t}\right)$$

$\mathrm{tf}_{t,c}$ = count of t in class c, $|c|$ = words in c, A = average words per class, f_t = count of t in all classes. Words in more than 15% of texts and generic words are removed.

Why this tool

Ordinary TF-IDF scores words within one document; here the unit is a whole class. A frequent word in one class but rare elsewhere scores high: it is what is *new* or *specific*.

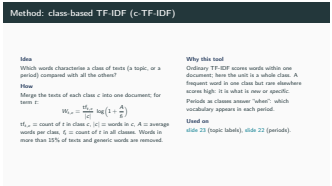
Periods as classes answer "when": which vocabulary appears in each period.

Used on

slide 23 (topic labels), slide 22 (periods).

Slide 45

Appendix: shown only if asked (0:00 in the budget). Whole topic (or period) treated as one document: a word frequent in one class and rare elsewhere scores high.



Method: covariate-adjusted logit

Idea

A raw change between periods can come from a change in *who* writes (more central banks, more Russian texts). Compare like with like.

How

For each outcome $y_t \in \{0, 1\}$ (a motivation, a topic):

$$P(y_t = 1) = \Lambda\big(a + b_{\text{period}} + c_{\text{actor}} + e \cdot \text{English}\big)$$

- **Average marginal effect**: predict every text in 2023–26, then in 2014–19, average the difference (points).
- 95% CI: robust (HC1) standard errors; coefficients drawn 400 times from their estimated distribution.

Why this tool

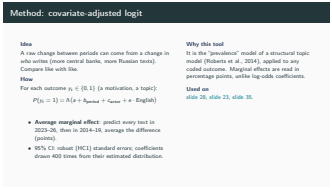
It is the "prevalence" model of a structural topic model (Roberts et al., 2014), applied to any coded outcome. Marginal effects are read in percentage points, unlike log-odds coefficients.

Used on

slide 28, slide 23, slide 35.

Slide 46

Appendix: shown only if asked (0:00 in the budget). Why: the mix of authors changes over time; the marginal effect compares the same texts in two periods, in points.



Method: codebook and validation

Idea

Write down, for each of 10 motivations, the phrases authors use to state it; flag a text when its introduction contains one.

How

1. Read 100 introductions (**development set**); write regular expressions in 7 languages.
2. Code 74 *other* texts by reading (**test set**), never used for tuning; scores on the 50 written by economists.
3. Compare: precision $= \frac{TP}{TP+FP}$, recall $= \frac{TP}{TP+FN}$, Cohen's $\kappa = \frac{p_o - p_e}{1 - p_e}$ (agreement beyond chance).

Why this tool

Motivations need reading; a codebook makes the reading explicit, reproducible and checkable.

Five machine coders were tested on the same texts (SBERT $\times$ 2, NLI mDeBERTa, Qwen2.5 3B and 7B): see slide 29.

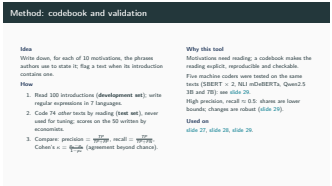
High precision, recall ≈ 0.5: shares are lower bounds; changes are robust (slide 29).

Used on

slide 27, slide 28, slide 29.

Slide 47

Appendix: shown only if asked (0:00 in the budget). Precision = share of codebook flags that are right; recall = share of read motivations found; kappa = agreement beyond chance.



Method: tone (FinBERT)

Idea

Measure how positive or negative the sentences *about CBDCs* are.

How

1. Keep the English sentences that name a CBDC (same term list as the filter).
2. FinBERT (a BERT model trained on financial news) gives $P(+), P(0), P(-)$ for each sentence.
3. Sentence tone $= P(+) - P(-) \in [-1, 1]$; averaged per text, then per group and year.
4. Validation: 90 sentences from economists' texts read blind; $\kappa = 0.52$ (all 0.53).

Why this tool

Dictionary methods (Loughran–McDonald) miss context ("not risky"); general sentiment models misread finance ("liability" is not negative). FinBERT is the standard for economic text (Araci, 2019).

Scoring only CBDC sentences isolates the tone *about CBDCs*, not about the economy.

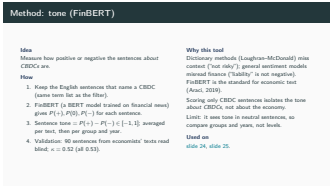
Limit: it sees tone in neutral sentences, so compare groups and years, not levels.

Used on

slide 24, slide 25.

Slide 48

Appendix: shown only if asked (0:00 in the budget). Why FinBERT: dictionaries miss context (“not risky”); general sentiment models misread finance (“liability” is not negative). Limit: sees tone in neutral sentences, so compare, do not read levels.



Method: stance (reading + local language model)

Idea

Tone is about words; stance is about the *author's position*: favourable, neutral or critical towards issuing a CBDC.

How

- **Levels**: texts read by hand per actor; share with a Wilson interval $\frac{\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}}{1+z^2/n}$.
- **Changes**: Qwen2.5 7B, run locally, labels a random sample of up to 100 texts per actor × period; change $= \hat{p}_{\text{late}} - \hat{p}_{\text{early}}$, CI $\pm 1.96\sqrt{\sum \hat{p}(1-\hat{p})/n}$.

Why this tool

Stance needs understanding of the argument: only a language model can do it at scale. Four coders were tested (slide 56); the LLM is best ($\kappa = 0.45$) but over-calls "favourable".

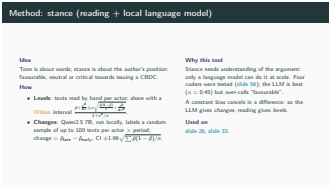
A constant bias cancels in a difference: so the LLM gives *changes*, reading gives *levels*.

Used on

slide 26, slide 33.

Slide 49

Appendix: shown only if asked (0:00 in the budget). Levels from reading (Wilson CI); changes from the LLM, because a constant bias cancels in a difference.



Method: bursts (Kleinberg)

Idea

Let the data say when a keyword suddenly becomes frequent, instead of choosing dates.

How

For keyword w, quarter t: r_t of d_t papers use it.

- Two states: normal rate $p_0 = R/D$ (overall), burst rate $p_1 = 2p_0$.
- Cost of a quarter in state i: $\sigma_i(t) = -\ln\left[\binom{d_t}{r_t} p_i^{r_t}(1-p_i)^{d_t - r_t}\right]$; entering a burst costs $\ln T$.
- The cheapest sequence of states (Viterbi algorithm) gives the bursts; weight = cost saved.

Why this tool

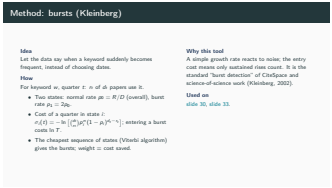
A simple growth rate reacts to noise; the entry cost means only sustained rises count. It is the standard "burst detection" of CiteSpace and science-of-science work (Kleinberg, 2002).

Used on

slide 30, slide 33.

Slide 50

Appendix: shown only if asked (0:00 in the budget). The entry cost means only sustained rises count; a simple growth rate would react to noise.



Method: event study with placebo dates

Idea

Does writing on CBDCs jump right after a policy event, more than on an ordinary day?

How

- Baseline rate λ = texts per day over days $[-90, -15]$ before the event.
- Expected over $[0, 14]$: $E = 15\lambda$; ratio $= N/E$; $z = (N - E)/\sqrt{E}$.
- Placebo: the same z on 500 random dates; p = share of random dates with a larger z.

Why this tool

The usual Poisson test assumes a stable rate; attention trends upward and comes in waves, so 13% of random dates look "significant". Comparing with random dates gives the right yardstick (as in finance event studies, MacKinlay 1997).

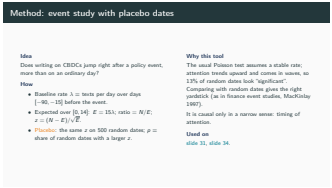
It is causal only in a narrow sense: timing of attention.

Used on

slide 31, slide 34.

Slide 51

Appendix: shown only if asked (0:00 in the budget). The naive Poisson test flags 13% of random dates: placebo dates give the right yardstick. Causal only for the timing of attention.



Method: do events move the tone?

Idea

Same logic as the event study, applied to the tone of the texts rather than their number.

How

- Texts dated to the day, with a tone score.
- Δ = mean tone over $[t_0, t_0 + 180)$ minus mean tone over $[t_0 - 180, t_0)$.
- Same Δ on 500 random dates; p = share with $|\Delta|$ at least as large (two-sided).

Why this tool

Tone rises over time anyway; a before/after difference alone would credit every event with that trend. Random dates carry the same trend, so only a jump beyond it counts.

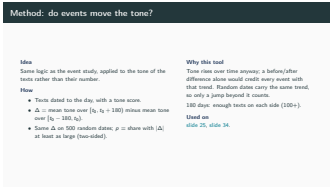
180 days: enough texts on each side (100+).

Used on

slide 25, slide 34.

Slide 52

Appendix: shown only if asked (0:00 in the budget). Random dates carry the same upward trend in tone, so only a jump beyond the trend counts.



Method: difference-in-differences

Idea

If the 2022 sanctions pushed economists to discuss geopolitics, texts about Russia should change *more* than the others after 2022.

How

Linear probability model, robust SEs; y = motivation stated, T = text about Russia or in Russian:

$$y_{it} = a_T + b_t + \sum_{k \neq 2021} \gamma_k \, T_i \, 1[t = k] + \varepsilon_{it}$$

γ_k = gap in year k relative to 2021. Pooled: one γ for $t > 2021$.

Why this tool

A simple before/after mixes the sanctions with everything else that happened in 2022; the other texts serve as the control.

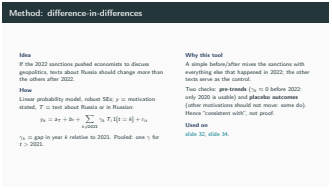
Two checks: **pre-trends** ($\gamma_k \approx 0$ before 2022: only 2020 is usable) and **placebo outcomes** (other motivations should not move: some do). Hence "consistent with", not proof.

Used on

slide 32, slide 34.

Slide 53

Appendix: shown only if asked (0:00 in the budget). Checks: pre-trend (only 2020 usable) and placebo outcomes (other motivations should not move; some do). Hence "consistent with".



Method: robustness

Idea

Re-run the headline results on samples that remove one possible source of error at a time.

How

- **Main theme only**: the secondary theme label is noisy.
- **English only**: language mix changes over time.
- **Journal articles only**: working papers and chapters differ.
- **Without the current year**: partial year.
- **High-recall coding**: codebook + SBERT matches.
- **All disciplines**: the economist filter itself.

Why this tool

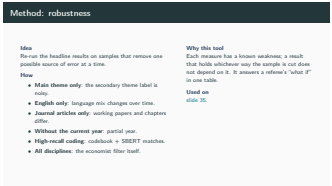
Each measure has a known weakness; a result that holds whichever way the sample is cut does not depend on it. It answers a referee's "what if" in one table.

Used on

slide 35.

Slide 54

Appendix: shown only if asked (0:00 in the budget). A result that holds however the sample is cut does not depend on that choice.



Backup

Backup: validation of the motivation coding

Codebook tuned on 100 texts (development set), then evaluated on the 50 held-out texts written by economists (of 74); unit = first 1,500 characters of the introduction / abstract.

Motivation	Read	Coded	Precision	Recall	κ
Privacy and surveillance	3	2	1.00	0.67	0.79
Geopolitics and sanctions	7	4	1.00	0.57	0.70
Technology (DLT, programmability)	11	11	0.73	0.73	0.65
Financial inclusion	7	3	1.00	0.43	0.56
Monetary policy	7	9	0.56	0.71	0.55
Cross-border payments	1	3	0.33	1.00	0.48
Financial stability	10	7	0.57	0.40	0.37
Monetary sovereignty	6	1	1.00	0.17	0.26
Decline of cash	7	1	1.00	0.14	0.22
Payment efficiency	16	3	0.67	0.12	0.12
All	75	44	0.70	0.41	0.46

Read = texts where the reading found the motivation; Coded = texts flagged by the codebook. Files: `presentation/validation_motivations_dev.csv`, `_test.csv`.

Slide 55

Appendix: shown only if asked (0:00 in the budget). Weakest: payment efficiency (said in many different ways) and decline of cash; strongest: privacy, geopolitics, technology.

Backup: validation of the motivation coding

Codebook tuned on 100 texts (development set), then evaluated on the 50 held-out texts written by economists (of 74); unit = first 1,500 characters of the introduction / abstract.

Motivation	Read	Coded	Precision	Recall	κ
Privacy and surveillance	3	2	1.00	0.67	0.79
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Financial inclusion	7	3	1.00	0.43	0.56
Monetary policy	7	9	0.56	0.71	0.55
Cross-border payments	1	3	0.33	1.00	0.48
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Payment efficiency	16	3	0.67	0.12	0.12
All	75	44	0.70	0.41	0.46

Read = texts where the reading found the motivation; Coded = texts flagged by the codebook. Files: presentation/validation_motivations_dev.csv, _test.csv.

Backup: validation of the stance coders

Each coder against the stance read by hand (favourable / neutral / critical); thresholds and classifiers by 5-fold cross-validation.

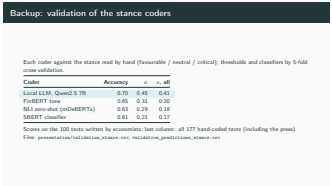
Coder	**Accuracy**	κ	κ, **all**
Local LLM, Qwen2.5 7B	0.70	0.45	0.41
FinBERT tone	0.65	0.31	0.20
NLI zero-shot (mDeBERTa)	0.63	0.29	0.18
SBERT classifier	0.61	0.21	0.17

Scores on the 100 texts written by economists; last column: all 177 hand-coded texts (including the press).

Files: `presentation/validation_stance.csv`, `validation_predictions_stance.csv`.

Slide 56

Appendix: shown only if asked (0:00 in the budget). The LLM is best (kappa 0.45 on economists' texts); FinBERT tone is a poor proxy for stance (0.31).



Backup: validation of the tone measure

Sentences sampled per FinBERT label, read blind to the label; table: the 90 from economists' texts (30 per label).

FinBERT ↓ / reading →	**negative**	**neutral**	**positive**	**Precision**
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

File: `presentation/validation_tone.csv`.

Slide 57

Appendix: shown only if asked (0:00 in the budget). Most errors are neutral sentences called positive or negative; positive vs negative confused only 3 times in 90.

Backup: validation of the tone measure

Sentences sampled per FinBERT label, read blind to the label; table: the 90 from economists' texts (30 per label).

FinBERT ↓ / reading →	negative	neutral	positive	Precision
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts read blind (of 136); agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

File: `presentation/validation_tone.csv`.

Backup: how the relevance score behaves

Example	Score	Label
Paper, CBDC in title	87	main theme
Paper, generic title, CBDC abstract	80	main theme
Paper, one passing mention	11	mentioned only
Speech, one paragraph on it	18	mentioned only
Speech devoted to it	80	main theme
Medical case report	0	off-topic

Slide 58

Appendix: shown only if asked (0:00 in the budget). CBDC in title: 87; generic title but CBDC abstract: 80; one passing mention: 11; medical case report: 0.

Backup: how the relevance score behaves

Example	Score	Label
Paper, CBDC in title	87	main theme
Paper, generic title, CBDC abstract	80	main theme
Paper, one passing mention	11	mentioned only
Speech, one paragraph on it	18	mentioned only
Speech devoted to it	80	main theme
Medical case report	0	off-topic

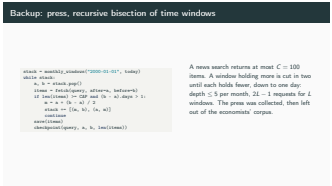
Backup: press, recursive bisection of time windows

```
stack = monthly_windows("2000-01-01", today)
while stack:
    a, b = stack.pop()
    items = fetch(query, after=a, before=b)
    if len(items) >= CAP and (b - a).days > 1:
        m = a + (b - a) / 2
        stack += [(m, b), (a, m)]
        continue
    save(items)
    checkpoint(query, a, b, len(items))
```

A news search returns at most $C = 100$ items. A window holding more is cut in two until each holds fewer, down to one day: depth ≤ 5 per month, $2L - 1$ requests for L windows. The press was collected, then left out of the economists' corpus.

Slide 59

Appendix: shown only if asked (0:00 in the budget). The press was collected this way, then left out of the economists' corpus.



Backup: academic sources, cursor pagination under a rate limit

```
def iter_phrase(phrase):
    cursor = "*"
    while cursor:
        d = get(f'{API}?filter=title_and_abstract'
                f'.search:"{phrase}"&per-page=200'
                f'&cursor={cursor}')
        yield from d["results"]
        cursor = d["meta"].get("next_cursor")

class Throttle:
    def wait(self):
        dt = self.interval - (now() - self.last)
        if dt > 0: sleep(dt)
        self.last = now()
```

- Each phrase searched in titles and abstracts; results merged on the source id.
- Requests spaced by $t_{k+1} - t_k \geq \Delta$.
- On HTTP 429: back-off $w_j = \min(60j,\ 900)$ s.
- Reference lists resolved into citations (for co-citation).

Slide 60

Appendix: shown only if asked (0:00 in the budget). Each phrase searched in titles and abstracts; requests spaced to respect rate limits.

Backup: academic sources, cursor pagination under a rate limit

```
def iter_phrase(phrase):
    cursor = "*"
    while cursor:
        d = get("/API?title+title_and_abstract"
                "&search:"{phrase}"&per-page=200"
                f"&cursor={cursor}")
        yield from d["results"]
        cursor = d["meta"].get("next_cursor")

class Throttle:
    def wait(self):
        dt = self.interval - (now() - self.last)
        if dt > 0: sleep(dt)
        self.last = now()
```

- Each phrase searched in titles and abstracts; results merged on the source id.
- Requests spaced by $t_{i+1} - t_i \geq \Delta$.
- On HTTP 429: back-off $w_i = \min(6i),\ 900)$ s.
- Reference lists resolved into citations (for co-citation).

Backup: full text and de-duplication

Full text

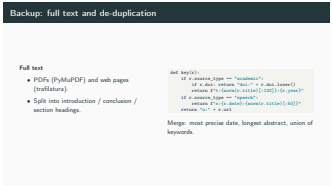
- PDFs (PyMuPDF) and web pages (trafilatura).
- Split into introduction / conclusion / section headings.

```
def key(r):
    if r.source_type == "academic":
        if r.doi: return "doi:" + r.doi.lower()
        return f"t:{norm(r.title)[:120]}:{r.year}"
    if r.source_type == "speech":
        return f"s:{r.date}:{norm(r.title)[:50]}"
    return "u:" + r.url
```

Merge: most precise date, longest abstract, union of keywords.

Slide 61

Appendix: shown only if asked (0:00 in the budget). Duplicates merged keeping the most precise date, the longest abstract and all keywords.



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Appendix: shown only if asked (0:00 in the budget).

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